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Final Project Proposal
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Do Lifestyle and Work Habits Predict Longevity? A Machine Learning Approach

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Problem Statement

“Work-life balance,” first coined in 1986, has received substantial attention from public health research in recent years, what the WHO and APA call the stress epidemic.¹ Research on the health effects of stress and burnout among workers has fueled initiatives for workers reasserting their personal time.² Many factors such as age, occupation, and daily time allocation (across work, rest, sleep, and exercise) may influence a person’s predicted lifespan.

In this final project, we aim to build a regression model that can accurately predict an individual’s expected longevity based on their lifestyle and gender using *Work-Life Balance and Longevity Dataset*.³ From this model, we will hopefully attain a better understanding of which lifestyle factors are most closely correlated with predicted lifespan.

General Approach

We will formulate the primary task as a supervised regression problem, where the goal is to predict an individual’s age at death based on lifestyle and work related input features. We will do this via linear regression in order to understand the relationship between individual features (such as hours of sleep, exercise, and work) and predicted lifespan (a real-valued target). Time permitting, we might also explore categorizing individuals into different longevity groups such as short, medium, and long lifespan based on their lifestyle characteristics. We could also apply unsupervised learning methods to identify groups of individuals with similar lifestyle patterns and analyze how those clusters differ in terms of longevity. One challenge might be handling extreme outliers, which could affect model performance.

Data

We plan on using the public, synthetic dataset downloaded from Kaggle, called [*Work-Life Balance and Longevity Dataset*](#). It contains information in tabular form on 10,000 individuals, specifically their gender, occupation type (divided into fourteen categories), daily work hours, daily rest hours, daily sleep hours, daily exercise hours, and age at death. We will treat the latter as our ground truth label and the rest as the feature vector.

¹ Stress, the “health epidemic of the 21st century.” (2019, May 1). HCA Healthcare Today. <https://hcahealthcaretoday.com/2019/04/30/stress-the-health-epidemic-of-the-21st-century/>

² Patricia Duchene. (2019, April 29). Beyond Work-Life Balance. Forbes. <https://www.forbes.com/sites/patriciaduchene/2019/04/29/beyond-work-life-balance/>

³ Oluwatosin Adewale & Tonic Fabricate. (n.d.). Work-Life Balance and Longevity Dataset [Dataset]. Kaggle. Retrieved March 5, 2026, from <https://www.kaggle.com/datasets/oluwatosinadewale/quality-of-life-data>

This data is fit for our problem and goals as it is substantially large, allowing for an ideal and sufficiently distributed train-validation-test split. Additionally, most of the features and the target are continuous, real values, making the dataset well-suited for a linear regression problem.

Measure of Success

For the regression task, where the goal is to predict a continuous variable, we will evaluate our model's performance using standard regression metrics such as MSE and R^2 . We hope to achieve a low MSE which will indicate that the model's predictions are closer to the true values, and a high R^2 which would indicate that our model explains more of the variance in lifespan. For the classification task, we will evaluate performance using accuracy (proportion of correct predictions), precision and recall (using true and false positive counts), and F1 score. Ideally, we will create a model that achieves a higher testing accuracy than sklearn's built-in LinearRegression model. Success will also be measured by our ability to observe relationships between the explanatory variables and longevity.

Milestones and Deliverables

Milestone 1: *Dataset processing*. We will obtain and load the dataset. We will split the dataset into training, validation, and testing splits.

Milestone 2: *Find our baseline*. We will instantiate and train an sklearn LinearRegression model on the data. Then, we will record its performance on the test split and use that as the baseline for our own model's success.

Milestone 3: *Program our own model*. Write the core functions for updating weights, fitting, scoring, etc. Determine whether we need to make the training process more efficient by using Stochastic Gradient Descent or other methods.

Milestone 4: *Fine-tune hyperparameters*. Based on results, keep adjusting hyperparameters, such as learning rate and discount rate, until we reach our goal accuracy.

Milestone 5: *Complete final report*. Write our report, documenting our methodology, implementation, and results.

Works Cited

Oluwatosin Adewale & Tonic Fabricate. (n.d.). *Work-Life Balance and Longevity Dataset*

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